

MATH 410 - WINTER 2026 - HOMEWORK 4, MIDTERM REVIEW

Turn in five of the following problems. (You should also review the problems from Homeworks 1–3.)

1. Let $x_0 \in \mathbb{R}^2$ and $R > 0$. Let f be the characteristic function of the ball of radius R around x_0 .

Hint. First work out the case $x_0 = 0$. Then, work out a general formula for the Radon transform of the translate of a function.

2. Let $Tf(x) = \int_{-\infty}^x f(t) dt$. Show that T is a convolution operator by finding its convolution kernel. This means that T is also a Fourier multiplier operator. Find its symbol (i.e. a function m such that $T = \mathcal{F}^{-1}m\mathcal{F}$).

3. Show that if m is bounded, then the operator T_m defined by $T_m = \mathcal{F}^{-1}m\mathcal{F}$ is a bounded linear operator from L^2 to L^2 , with operator norm bounded by $\|m\|_{L^\infty}$. To be clear, the operator T_m is defined so that we have $\mathcal{F}[T_m f](\xi) = m(\xi)\mathcal{F}f(\xi)$.

4. Show that the unfiltered back-projection formula corresponds to convolution with $\frac{1}{|x|}$, i.e.

$$\frac{1}{2\pi} \int_0^{2\pi} \mathcal{R}f(\langle x, \omega \rangle, \omega) d\theta = \frac{1}{\pi} [f * k](x), \quad k(x) = \frac{1}{|x|}.$$

5. Let $f_a(x) = e^{-a|x|^2}$. Compute $f_a * f_b$, where $a, b > 0$. (*Hint:* Use the Fourier transform).

6. Suppose $g(x) = f(\lambda x)$ for some $\lambda > 0$. Show that $\hat{g}(\xi) = \lambda^{-1} \hat{f}(\frac{\xi}{\lambda})$.

7. Suppose $g(x) = f(x - x_0)$ for some $x_0 \in \mathbb{R}$. Show that $\hat{g}(\xi) = e^{-i\xi x_0} \hat{f}(\xi)$.

8. State the Fourier inversion formula for $f \in L^1$. Make sure to include the assumptions we need to guarantee that the formula holds pointwise.

9. State and prove the Filtered Back Projection formula.

10. Describe the algebraic approach to X-ray CT reconstruction, namely, how the problem of reconstruction can be reduced to a linear system of the form $p = rx$. Then describe the Kaczmarz iteration (also known as method of projections), i.e. the algorithm that can be used to look for an approximate solution.

11. I should have put this on Homework 3, but that homework was long enough already. Let $f = \chi_{[0,1]}(x)$. Compute $f * f$.

12. Let H be the characteristic function of $(0, \infty)$. Prove that $\hat{H}(\xi) = \frac{1}{i\xi}$ for all $\xi \neq 0$. Then compute the Fourier transform of the characteristic function of $(0, R)$ for general $R > 0$.

13. Let $\mathcal{F}$ denote the Fourier transform in n dimensions, and let Δ be the Laplacian operator, i.e.

$$\Delta f = \sum_{j=1}^n \frac{\partial^2 f}{\partial x_j^2}.$$

Prove that $\mathcal{F}[\Delta f](\xi) = -|\xi|^2 \mathcal{F}f(\xi)$. (If you are worried about working in n dimensions, just prove it for $n = 1$.)

14. (Required for 510; optional for 410) Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a compactly supported function that is infinitely differentiable everywhere except for $x = 0$, where it has a (finite) jump discontinuity. Prove that there exist $C_1, C_2 > 0$ such that

$$C_1 |\xi|^{-1} \leq |\hat{f}(\xi)| \leq C_2 |\xi|^{-1}$$

for all $\xi \geq 1$.